Thomas **FUZELLIER**

Embedded Systems Engineer

Embedded systems engineer (ENSEA) with over a year of hands-on experience, from STM32 firmware development in C/C++ to on-site commissioning and field diagnostics. Versatile profile spanning software and electronics, with additional experience in applied AI.

Experience

Embedded Systems Engineer – Consultant
Sep 2026 – Present **Ethic Opex - GCK** Cournon, France

After-sales assignment: vehicle troubleshooting, reliability and continuous improvement.

- **Diagnose hardware and software faults** reported by end users
- Implement **corrective actions**: software realignment, **calibration**, test instrumentation
- Analyse data with **network and CAN tools**; run **dynamic tests** to validate fixes
- Write **test reports** and service records; escalate field issues to the **design office**
- Deliver **training sessions** to partner repair workshops

Embedded Systems Engineer
Jan 2025 – Jan 2026 **Semi Lift** Toulon, France

Lift manufacturing SME bringing its electronics development in-house.

- Developed and maintained **C/C++ lift control firmware** on **STM32 microcontrollers**
- Developed **drivers** to interface sensors and actuators
- **Integrated and validated** sensors on lift systems
- **Commissioned installations on site** and carried out troubleshooting
- Built **Python tools** to automate testing and analyse operating data

NLP / LLM Engineering Intern – Healthcare
May – Dec 2024 **CTB - MEDAL Laboratory** University of Madrid, Spain

Automated structuring of confidential Spanish clinical notes using LLMs (Llama).

- Built an **end-to-end NLP pipeline** in Python: extraction, normalisation and structuring
- Integrated **Llama models via REST API**: prompt engineering, token handling, benchmarks
- Integrated a **knowledge graph** to link the extracted medical information
- **Productionised** the pipeline: modular architecture, unit tests, technical documentation

Embedded Systems & Instrumentation Engineering Intern
Jun – Aug 2023 **Stratolia** Palaiseau, France

Earth-observation payload carried by a stratospheric balloon.

- Developed a **ground control system** for an onboard camera
- Carried out the **mechanical design** and prototyping of the gondola
- Developed **embedded software** on a microcontroller for sensor data acquisition
- **Designed the electronics** and produced the **PCBs**
- Built a **test bench** simulating flight conditions

Education

MSc in Engineering
Sep 2020 – Dec 2024
ENSEA — École Nationale Supérieure de l'Électronique et de ses Applications, Cergy, France
Diplôme d'Ingénieur, Computer Science & Systems major

Preparatory Class (ATS)
Sep 2019 – Jun 2020
Lycée des Remparts, Marseille, France
Intensive mathematics and physics for engineering school entrance

Technical Degree (DUT)
Sep 2017 – Jun 2019
Aix-Marseille University, Marseille, France
Electrical Engineering and Industrial Computing (2-year degree)

Projects

AI-Assisted Job Application Generator
Personal project GitHub Python · Tkinter · Groq API (LLM) · pytest

Desktop app producing a tailored CV, PDF cover letter and Excel tracker from a job posting.

- **LLM pipeline**: language detection, selection of the best CV and pre-written paragraphs
- AI restricted to **selecting validated content**, with no generated text, for controlled output
- **Modular architecture**, **PDF** and **Excel** export, **unit tests** with a mocked API

Contact

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Based in France, open to relocation within France

Skills

Programming
C/C++, Python, Java, SQL

Embedded Systems
STM32, Arduino, CAN bus

Electronics
PCB design (KiCad), embedded electronics

AI / Machine Learning
LLMs (GPT, LLaMA), NLP, Transformers, Prompt Engineering

Web & APIs
REST APIs, HTTP, HTML, CSS, JavaScript

Tools
Git, Linux, Docker, Matlab, Fusion 360, PyTorch, Hugging Face, Pandas, NumPy

Languages

French
Native

English
Professional working proficiency (TOEIC 810)

Spanish
Intermediate

Interests

Woodworking
Handcrafted furniture

Games
Card, board and video games

Sport
Street lifting, tennis